

Solivia Commercial *Decision Support System.*

Pricing, procurement, distributor-opportunity and supply-risk analytics for a healthcare distribution startup.

3

planned phases

20

defined steps

98 to 157

estimated project hours

These totals are calculated from the supplied plan: **36 to 56 hours** for Phase 1, **50 to 80** for Phase 2 and **12 to 21** for Phase 3. They are estimates, not logged work.



A blueprint with a working illustration, not a results claim.

The project plan supplies the scope and roadmap. The mathematical specification below is a proposed extension, and every numerical business example is **synthetic**. Real dataset counts, validated model performance and measured commercial benefits have not been supplied. The public scenario lab is implemented here; the full private commercial system is not claimed to be operational.

01 / EXPLAIN THE BUSINESS PROBLEM

A good quote is more than *a selling price.*

My central question is how a healthcare distribution startup can make more defensible **pricing, quotation and procurement decisions** when supplier costs, freight, demand and delivery timing are uncertain.

The proposed users are the people preparing quotations, comparing procurement alternatives and reviewing commercial opportunities. The system should make the cost basis, expected contribution, cash requirement and uncertainty visible beside each decision.

Scope boundary: this is a commercial decision-support project. It does not make clinical recommendations or use patient records.

Map the work before *claiming to improve it.*

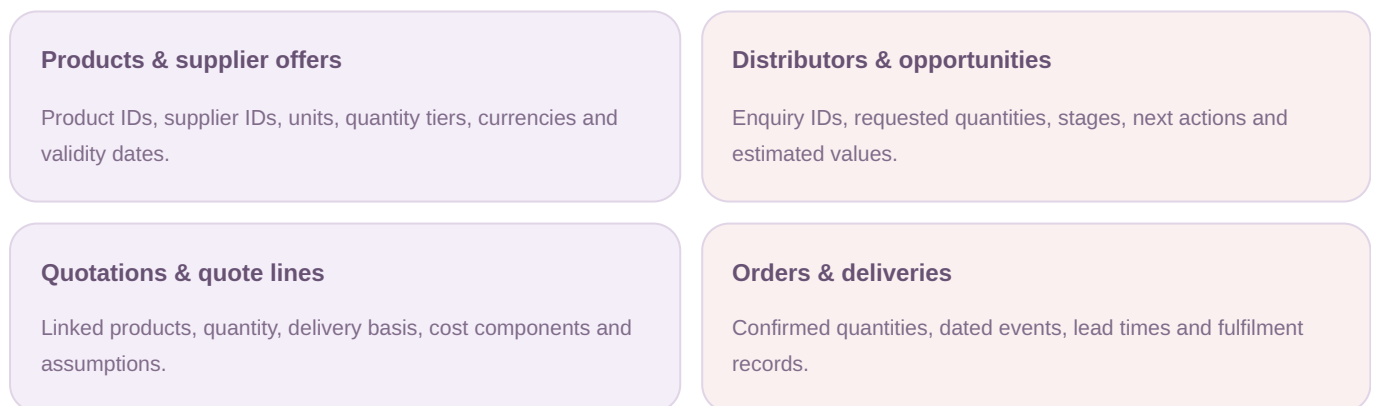
The first task is to document how supplier information, quotations, enquiries and purchasing decisions currently move through Solivia. I will use **stakeholder input and available records** to identify genuine difficulties.



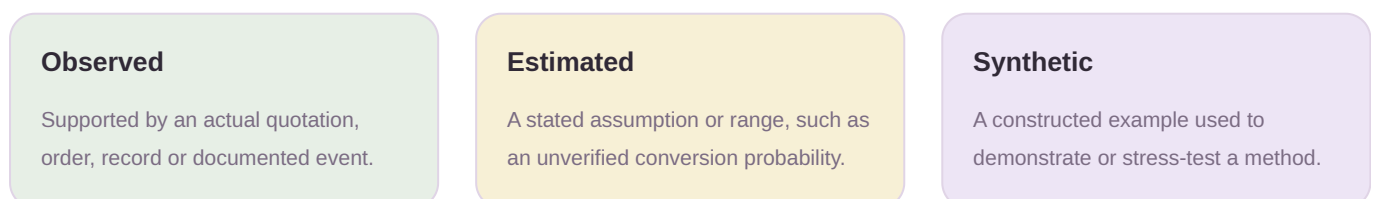
No baseline workflow study is attached to the plan. I therefore do not assume a particular spreadsheet setup, invent delays or assign an efficiency gain before measuring the existing process.

Give every number *a source.*

The plan calls for linked SQL records covering products, suppliers, prices, minimum order quantities, lead times, quotations and enquiries. A proposed relational structure could group the information as follows; this is a **design sketch, not a deployed database**.



Stable identifiers should link these records. Transaction lines should be separated from document headers, and **units, currencies, source references and dates** retained rather than inferred later.



What the data audit will quantify

Coverage Number of unique products, suppliers and completed orders; first and last usable dates; missing periods by product.	Completeness Required populated fields ÷ required field slots × 100%. A missing sale is not automatically a zero sale.
Integrity Duplicate primary keys, unmatched references, inconsistent units and records lacking a source or date.	Lead-time statistics Count, mean, median and spread of observed lead times, with the event definition and units stated.

These are proposed audit measures. The plan supplies no production row counts or observed summary statistics, so none are invented here.

04 / EXPLAIN THE DECISION METHODOLOGY

Make the reasoning *inspectable*.

The following equations turn the plan into a proposed quantitative specification. **They are methodological additions**, not evidence that the full model has been built. The browser lab implements the simplified quotation equations only.

A. Landed cost, quotation and contribution

For an order of q units, let c be the supplier cost per unit, x the exchange rate into SGD, F total freight and I total insurance. In the simplified demonstration:

Total included cost, in SGD

$$C(q) = q \times c \times x + F + I$$

Landed cost per unit

$$c_L = C(q) / q$$

Required price for target margin m

$$P^* = c_L / (1 - m)$$

Contribution and contribution margin

$$CM = qP - C(q)$$

$$\text{Margin} = CM / (qP)$$

Why divide by $(1 - m)$? Because margin is measured against revenue: $m = (P - c_L) / P$. Rearranging gives $P(1 - m) = c_L$.

Multiplying cost by $1 + m$ would instead apply a markup.

The lab rounds the required price **up to the nearest cent** to preserve the selected minimum margin. It uses that quoted price for revenue and contribution, with full-precision costs underneath.

Model boundary: quantity is positive and the target margin is below 100%. All units are assumed sold. Only separately payable freight and insurance should be entered, so delivery charges already included in a supplier price are not counted twice. Other variable costs, taxes, overhead and payment timing are not modelled in the public example. A live quotation would need those inputs checked.

B. MR: does selling more add enough value?

Marginal revenue (MR) is the change in total revenue per additional unit sold. For two discrete commercial scenarios, compare changes in total revenue and total relevant cost, not just the new price. [2]

$$MR = (R_2 - R_1) / (q_2 - q_1)$$

$$MC = (C_2 - C_1) / (q_2 - q_1)$$

$$\Delta CM = (R_2 - C_2) - (R_1 - C_1)$$

SYNTHETIC SCENARIO A

1,000 units at S\$20

Total revenue: **S\$20,000**

Total relevant cost: **S\$12,500**

Contribution: **S\$7,500**

SYNTHETIC SCENARIO B

1,200 units at S\$19

Total revenue: **S\$22,800**

Total relevant cost: **S\$14,400**

Contribution: **S\$8,400**

In this constructed comparison, **MR = S\$2,800 / 200 = S\$14** per additional unit, **MC = S\$1,900 / 200 = S\$9.50**, and contribution rises by **S\$900**. MR is below the new S\$19 price because the lower price applies to every unit in scenario B.

Quantities, prices and costs are assumptions, not Solivia observations. This is a finite scenario comparison, not an estimated demand curve or proof that extra demand exists. A continuous model could use $R(q) = qP(q)$ and $MR = dR/dq$, but that would require a justified demand specification.

C. Statistics: evaluate later observations

The source plan gates forecasting on data readiness. Where evidence is sufficient, compare simple baselines with a justified alternative using **later observations not used to fit the model**. For error $e_t = y_t - \hat{y}_t$: [3]

$$\text{MAE} = \Sigma |e_t| / n$$

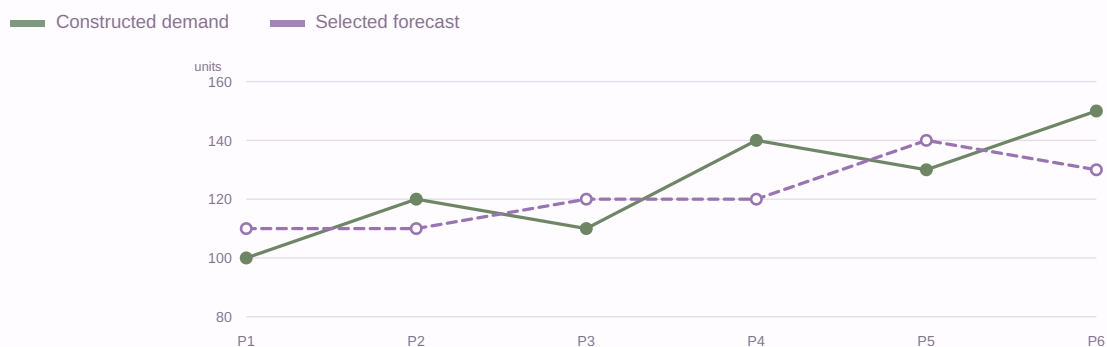
$$\text{RMSE} = \sqrt{(\Sigma e_t^2 / n)}$$

$$\text{Bias} = \Sigma e_t / n$$

MAE expresses average absolute error in the original units. RMSE gives large misses more influence. Positive bias under this sign convention means underprediction. Model choices should be made on training/validation data, with a final test reserved for evaluation.

Synthetic illustration only

Six periods. Two illustrative forecasts.



MAE

13.33 units

RMSE

14.14 units

Bias

+3.33 units

See every number behind the illustration

×

All six rows are synthetic, not a Solivia sales history.

Period	Constructed demand	Baseline	Candidate
1	100	110	105
2	120	110	118
3	110	120	115
4	140	120	132
5	130	140	134
6	150	130	143

For the baseline, absolute errors sum to 80 and squared errors sum to 1,200. Therefore $MAE = 80/6 = 13.33$ and $RMSE = \sqrt{(1,200/6)} = 14.14$. For the candidate, $MAE = 31/6 = 5.17$, $RMSE = \sqrt{(183/6)} = 5.52$ and bias = 0.50.

These calculations illustrate the measures. They do not establish the quality of a trained model or a realised improvement.

D. Inventory and commercial risk

A proposed inventory rule links the reorder point to demand over the replenishment lead time. In a simplified deterministic illustration:

$$ROP = \text{Expected lead-time demand} + \text{Safety stock}$$

Assume **40 units per day**, a **7-day** lead time and an explicitly assumed **60-unit buffer**. The illustrative reorder point is $40 \times 7 + 60 = 340$ units. The buffer is a scenario input, not an estimated service-level guarantee.

In the full project, safety stock should be justified by observed demand and lead-time uncertainty where possible. When the observations are insufficient, use labelled ranges rather than presenting chosen buffers as statistically estimated.

Sales concentration Distributor share = that distributor's confirmed sales ÷ total confirmed sales. Report pipeline shares separately; a zero denominator means the sales share is undefined.	Concentration index $HHI = \sum s_i^2$, using shares as fractions between 0 and 1. This proposed indicator summarises concentration; no risk thresholds are asserted.
Weighted pipeline $\Sigma(\text{probability} \times \text{opportunity value})$. Every conversion probability is an assumption unless supported by evidence. This is not booked revenue.	Peak cash exposure Maximum cumulative cash shortfall over the stated timeline. It requires payment dates and terms, not just a contribution-margin calculation.

E. Optimisation: choose a feasible order

The plan proposes constrained procurement optimisation. One possible **single-period deterministic model** is shown below, with zero starting inventory and fixed scenario prices and demand. It is a proposed specification, not an implemented solver.

For product i , let q_i be ordered units, s_i units sold and y_i a binary order indicator. Let p be selling price, c unit purchase cost, h additional holding cost per unsold unit, and k a fixed order cost.

Maximise scenario contribution after relevant inventory costs

$$\max \Sigma [p_i s_i - c_i q_i - h_i (q_i - s_i) - k_i y_i]$$

Subject to supported constraints

$$0 \leq s_i \leq d_i, \quad s_i \leq q_i$$

$$MOQ_i y_i \leq q_i \leq U_i y_i$$

$$\Sigma (c_i q_i + k_i y_i) \leq B$$

$$\Sigma v_i q_i \leq V$$

$$q_i, s_i \text{ are non-negative integers; } y_i \in \{0, 1\}$$

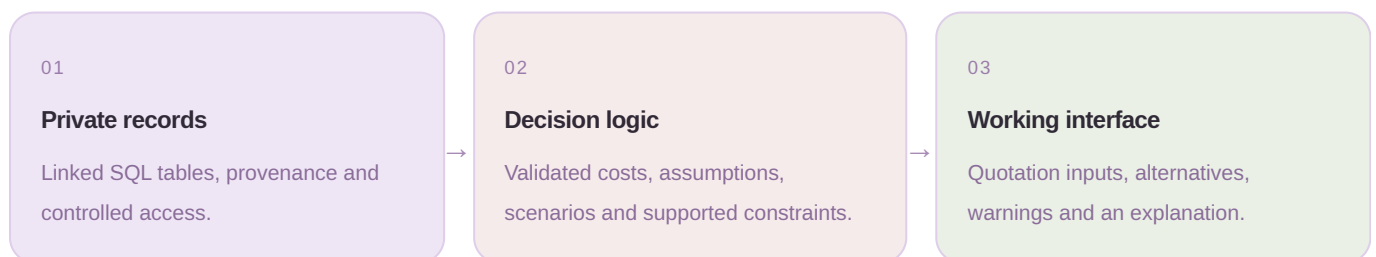
Here d is scenario demand, U a valid order upper bound, B the cash budget, v storage per unit and V total storage capacity. Purchase costs already cover all ordered units; h adds holding cost rather than charging for unsold units a second time.

Under optional purchasing, the zero-order solution is feasible. A “no feasible solution” test only makes sense once supported mandatory commitments or other incompatible constraints are introduced. The application should explain such conflicts rather than output an impossible order.

The full evaluation would compare the optimiser with simpler ordering rules under the **same scenarios**, examining contribution, stockouts, excess inventory and cash exposure. A stochastic extension or Monte Carlo stress test would require explicit probability assumptions.

05 / DESCRIBE THE SYSTEM DESIGN

Keep the logic connected.
Keep private data private.



The architecture above is the intended commercial workflow. The **public portfolio runs separately**: its quotation lab uses only the inputs shown below, and its forecast chart uses fixed synthetic arrays. Neither connects to company systems.

Implemented in this website

Cross-page search, the simplified quotation calculator, input validation and a synthetic forecast-error visualisation.

Planned for the full project

The private relational database, stakeholder-validated quotation workflow, forecasting where data permits, and a procurement optimiser.

06 / DEMONSTRATE A COMPLETE COMMERCIAL SCENARIO

A small lab for “*what changes when...?*”

Change the assumptions to see the arithmetic update. This is a **synthetic educational scenario**, not a live Solivia quotation or a forecast of demand.

Synthetic inputs · calculated outputs

Order quantity

Units

Supplier cost per unit

Supplier-currency units

Exchange rate

SGD per supplier-currency unit

Freight per order

SGD

Insurance per order

SGD

Target contribution margin

Percent of revenue

Available cash budget

SGD

Minimum quoted unit price

S\$16.67

Rounded up to the nearest cent

Landed unit cost

S\$12.50

Contribution

S\$4,170.00

Total included cost

S\$12,500.00

Actual quoted margin

25.01%

Illustrative revenue

S\$16,670.00

Cash budget headroom

S\$2,500.00

■ Procurement ■ Freight ■ Insurance ■ Contribution

Budget check: the included upfront cost fits within the illustrative budget, leaving S\$2,500.00.

All included costs are assumed payable upfront; no customer deposit is assumed. Cash headroom is therefore budget minus included cost. This is not a full working-capital timeline. Other costs are excluded, not silently estimated.

Read the default scenario from start to finish

Inputs: 1,000 units, supplier cost 12 per unit, exchange rate 1.00, freight S\$400, insurance S\$100 and target margin 25%.

Calculation: $C = 1,000 \times 12 \times 1 + 400 + 100 = \text{S\$12,500}$. Cost per unit is S\$12.50. Required unrounded price is $12.50 / 0.75 = \text{S\$16.666...}$; the rounded-up quotation is **S\$16.67**.

Output: revenue S\$16,670, contribution S\$4,170 and actual quoted margin **25.015%** before display rounding. With a S\$15,000 budget, the included upfront cost leaves **S\$2,500** of headroom.

Sensitivity: increase freight by S\$200 while leaving all other inputs unchanged. Total included cost becomes S\$12,700 and the new minimum price is **S\$16.94**. At the original S\$16.67 price, contribution would instead fall by S\$200 and margin to about **23.82%**. An input assumption changes the required quote.

All business inputs in this walkthrough are constructed for demonstration. The dynamic lab may show different values after editing; this walkthrough always describes the stated defaults.

07 / EVALUATE RESULTS AND BUSINESS VALUE

Measure what changed. *Do not guess the benefit.*

No measured commercial outcomes are provided in the source plan. The completed case study should report what testing actually establishes, with the denominator, test set and assumptions visible.

01

Calculation checks

Pass rate = passed cases / tested cases. Compare the engine with hand calculations, boundary cases, missing inputs and supplier-tier transitions. State the rounding rule and tolerance.

02

Forecast evaluation

Report **MAE, RMSE and bias** on later observations. Include sample size, forecast horizon and baseline performance. Defer claims where the series cannot support evaluation.

03

Decision quality

Compare contribution, stockouts, excess inventory and cash exposure under identical scenarios. Keep simulated improvements distinct from observed outcomes.

04

Workflow evidence

Record stakeholder feedback and, where tested, comparable task times. **Time change = (baseline – system time) / baseline × 100%**, with sample size and task conditions stated.

Current evidence status: the plan defines intended work. Website demonstrations illustrate calculations. Neither establishes increased revenue, realised cost savings, a production forecast or validated operational improvements.

08 / DISCUSS LIMITATIONS AND FINISH THE REPORT

The limits belong *beside the result.*

Thin sales history, missing dates, inconsistent product definitions and uncertain lead times may limit forecasting. An optimiser can only choose within its stated objective and constraints; a numerically attractive order is not useful when important business restrictions are absent.

The public examples omit taxes, other variable costs, credit terms, overhead and interactions between products. A synthetic dataset can demonstrate logic, but **cannot validate Solivia's actual demand or commercial performance.**

The final report should state what is operational versus experimental, document assumptions and sources, show selected anonymised workflows, and link to reproducible code only when the relevant repository is ready. Until then, this site links to **my GitHub profile**, not an invented project repository.

[My GitHub profile ↗](#)

[Methods I'm learning ↗](#)

THE ORIGINAL PLAN, OPENED UP

Three phases. *Twenty deliberate steps.*

The descriptions, rigour levels and time ranges below follow the supplied six-page plan. Punctuation and layout have been adapted for the web. **No step is marked complete without evidence.**

01

PHASE 1 · 6 STEPS · 36 TO 56 HOURS

Working commercial system

Organise the evidence, make the calculations transparent, then connect the workflow.

1.1 Create the product and supplier database

Inventory the information Solivia actually possesses, then organise products, suppliers, prices, minimum order quantities, lead times, quotations and enquiries into linked SQL tables. Record units, currencies, sources and dates. Separate observed information from estimates, and create a sanitised dataset for the public demonstration.

Hard rigour

8 to 12 hours

1.2 Build the landed-cost and margin calculator

Calculate per-unit procurement, freight, insurance and other applicable variable costs without double-counting them. Calculate contribution margin and cash required. Document the assumptions behind each calculation, and manually check several examples against the system's output.

Medium rigour

5 to 8 hours

1.3 Build the quotation scenario engine

Allow a user to select a product, quantity, destination, delivery basis and target margin. Return a calculated quotation, cost breakdown and expected contribution. Compare supplier pricing tiers and order quantities, flag missing inputs and show when a quotation fails the selected margin requirement.

Hard rigour

5 to 8 hours

1.4 Create the distributor and tender-opportunity tracker

Track each enquiry through quotation, documentation, purchase order, production and delivery stages. Include requested quantities, next actions, estimated deal value and expected timing. Clearly distinguish confirmed orders from opportunities; label conversion probabilities as assumptions rather than established facts.

Medium rigour

5 to 8 hours

1.5 Build the commercial-risk dashboard

Show supplier, distributor and product concentration, tender dependence, working-capital exposure and outstanding documentation. Define each indicator and explain what it measures. Keep actual sales concentration separate from projected pipeline concentration so the dashboard does not imply revenue that has not occurred.

Medium rigour

5 to 8 hours

1.6 Connect everything in an interactive scenario laboratory

Let users adjust quantities, supplier prices, freight, exchange rates and margins, then see the effects on quotations, contribution and cash requirements. Test the connected workflow with a Solivia stakeholder, check invalid inputs and correct usability problems. Keep confidential information out of the public demonstration.

Hard rigour

8 to 12 hours

02

PHASE 2 · 6 STEPS · 50 TO 80 HOURS

Forecasting and procurement optimisation

Only forecast when the data supports evaluation. Compare decisions, not just models.

2.1 Decide whether the data is ready for forecasting

Review completed orders, available history, missing periods, product consistency and lead-time records. Decide which products can support genuine forecast evaluation. Where the evidence is insufficient, defer forecasting or use an explicitly labelled simulation, not a claimed prediction of Solivia's sales.

Medium rigour

4 to 6 hours

2.2 Build and compare demand forecasts

Start with simple baselines, such as previous-period demand or a moving average. Add a more complex method only where justified. Evaluate predictions using later observations that were not used to fit the model, examine forecast errors and bias, and explain the model-selection decision.

Hard rigour

10 to 16 hours

2.3 Translate uncertainty into inventory rules

Estimate plausible demand and lead-time uncertainty. Use this to compare reorder points and safety-stock policies. Explain the trade-off between availability, excess stock and tied-up cash. Distinguish uncertainty estimated from observations from ranges chosen purely for scenario testing.

Hard rigour

8 to 12 hours

2.4 Build the constrained procurement optimiser

Define order quantities as decision variables and choose a clearly stated objective, such as expected contribution after relevant inventory costs. Add supported constraints: budget, minimum order quantities, storage or supplier restrictions. Test whether the recommendations are feasible and whether the model handles cases with no feasible solution.

Hard rigour

12 to 20 hours

2.5 Stress-test the procurement policies

Compare the optimiser against simpler ordering rules under the same demand and supplier scenarios. Use scenario analysis and, where justified, Monte Carlo simulation to vary demand, delays, freight and tender outcomes. Compare contribution, stockouts, excess inventory and cash exposure, not just forecast accuracy.

Hard rigour

10 to 16 hours

2.6 Integrate and validate the recommendations

Add forecasts, inventory rules and procurement comparisons to the application. Show assumptions and uncertainty beside each recommendation. Review results with a Solivia stakeholder, investigate unrealistic outputs and record changes. Distinguish simulated improvements from any benefits observed during actual use.

Hard rigour

6 to 10 hours

03

Written technical case study

PHASE 3 · 8 STEPS · 12 TO 21 HOURS

Make the reasoning reproducible and keep every conclusion proportional to the evidence.

3.1 Explain the business problem

State which pricing, quotation or procurement decisions the system supports, who will use it and what the project aims to improve. Define the boundaries clearly: this is a commercial decision-support tool, not a clinical recommendation system.

Easy rigour

1 to 2 hours

3.2 Describe the existing workflow

Explain how Solivia currently handles supplier information, quotations, enquiries and purchasing decisions. Use stakeholder input and available records to identify genuine difficulties. Do not invent inefficiencies simply to make the new system appear useful.

Medium rigour

1 to 2 hours

3.3 Document the data design

Explain the database structure, relationships, sources, cleaning decisions and confidentiality measures. Include a compact schema diagram or data dictionary. Identify which fields are observed, estimated or simulated and explain any important missing information.

Medium rigour

1 to 2 hours

3.4 Explain the decision methodology

Present the important cost and margin calculations, assumptions and scenario logic. Include forecasting, inventory and optimisation methods only if implemented. Reference methodological sources and justify why the chosen methods fit the available data and business problem.

Hard rigour

2 to 3 hours

3.5 Describe the system design

Show how the database, calculation engine and interface connect. Explain the main user workflow, input checks and separation between private company data and the public demonstration. Include a readable architecture diagram and selected screenshots.

Medium rigour

1 to 2 hours

3.6 Demonstrate one complete commercial scenario

Walk through an anonymised quotation or procurement decision: inputs → calculations → alternatives → output → interpretation. Explain how changing one important assumption changes the decision. Identify clearly whether the example is real, anonymised or synthetic.

Medium rigour

2 to 3 hours

3.7 Evaluate results and business value

Report what testing actually established: calculation accuracy, consistent quotations, clearer comparisons, stakeholder feedback or measured task-time changes. Where Phase 2 is completed, report comparisons against baseline policies. Do not claim revenue gains or cost savings without supporting evidence.

Hard rigour

2 to 3 hours

3.8 Discuss limitations and finish the report

Explain missing data, uncertain assumptions, model weaknesses and remaining development needs. State what is operational versus experimental. Finalise references, captions and reproducibility links; check that every conclusion matches the evidence and that the report contains no confidential information.

Medium rigour

2 to 4 hours

Where the content *comes from.*

1. **Project scope and roadmap:** *Passion Project Plan - Google Docs.pdf*, supplied by Sahana Sharma, six pages. Phase 1: pages 1 and 2. Phase 2: pages 3 and 4. Phase 3: pages 5 and 6. Time totals are calculated by summing the supplied ranges.

2. **Marginal-revenue definition:** OpenStax, *Principles of Economics 3e*, section 8.2.

[Read the methodological reference ↗](#)

3. **Forecast-error measures and held-out evaluation:** Rob J Hyndman and George Athanasopoulos, *Forecasting: Principles and Practice*, third edition, section 5.8.

[Read the methodological reference ↗](#)

The cost derivation, proposed optimisation specification, data-audit measures and synthetic examples are **additions for this portfolio write-up**. They do not appear as measured findings in the source plan. Existing biographical content and evidence elsewhere on the site come from the supplied portfolio and certificate files.

References checked 8 September 2026. The original certificates remain unmodified.